

AP[®] Physics 1: Algebra-Based Practice Exam

From the 2015 Administration

NOTE: This is a modified version of the 2015 AP Physics 1: Algebra-Based Exam.

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AP[®] PHYSICS 1 TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS

Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	Coulomb's law constant, $k = 1/4\pi\epsilon_0 = 9.0 \times 10^9$ N·m ² /C ²
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Universal gravitational constant, $G = 6.67 \times 10^{-11}$ m ³ /kg·s ²
Speed of light, $c = 3.00 \times 10^8$ m/s	Acceleration due to gravity at Earth's surface, $g = 9.8$ m/s ²

UNIT SYMBOLS	meter, m	kelvin, K	watt, W	degree Celsius, °C
	kilogram, kg	hertz, Hz	coulomb, C	
	second, s	newton, N	volt, V	
	ampere, A	joule, J	ohm, Ω	

PREFIXES		
Factor	Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	1/2	3/5	$\sqrt{2}/2$	4/5	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	4/5	$\sqrt{2}/2$	3/5	1/2	0
$\tan \theta$	0	$\sqrt{3}/3$	3/4	1	4/3	$\sqrt{3}$	∞

The following conventions are used in this exam.

- I. The frame of reference of any problem is assumed to be inertial unless otherwise stated.
- II. Assume air resistance is negligible unless otherwise stated.
- III. In all situations, positive work is defined as work done on a system.
- IV. The direction of current is conventional current: the direction in which positive charge would drift.
- V. Assume all batteries and meters are ideal unless otherwise stated.

AP[®] PHYSICS 1 EQUATIONS

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_n|$$

$$a_c = \frac{v^2}{r}$$

$$\vec{p} = m\vec{v}$$

$$\Delta \vec{p} = \vec{F} \Delta t$$

$$K = \frac{1}{2} m v^2$$

$$\Delta E = W = F_{\parallel} d = F d \cos \theta$$

$$P = \frac{\Delta E}{\Delta t}$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega = \omega_0 + \alpha t$$

$$x = A \cos(2\pi f t)$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$\tau = r_{\perp} F = r F \sin \theta$$

$$L = I \omega$$

$$\Delta L = \tau \Delta t$$

$$K = \frac{1}{2} I \omega^2$$

$$|\vec{F}_s| = k |\vec{x}|$$

$$U_s = \frac{1}{2} k x^2$$

$$\rho = \frac{m}{V}$$

a = acceleration
 A = amplitude
 d = distance
 E = energy
 f = frequency
 F = force
 I = rotational inertia
 K = kinetic energy
 k = spring constant
 L = angular momentum
 ℓ = length
 m = mass
 P = power
 p = momentum
 r = radius or separation
 T = period
 t = time
 U = potential energy
 V = volume
 v = speed
 W = work done on a system
 x = position
 y = height
 α = angular acceleration
 μ = coefficient of friction
 θ = angle
 ρ = density
 τ = torque
 ω = angular speed

$$\Delta U_g = m g \Delta y$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$|\vec{F}_g| = G \frac{m_1 m_2}{r^2}$$

$$\vec{g} = \frac{\vec{F}_g}{m}$$

$$U_G = -\frac{G m_1 m_2}{r}$$

ELECTRICITY

$$|\vec{F}_E| = k \left| \frac{q_1 q_2}{r^2} \right|$$

$$I = \frac{\Delta q}{\Delta t}$$

$$R = \frac{\rho \ell}{A}$$

$$I = \frac{\Delta V}{R}$$

$$P = I \Delta V$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

A = area
 F = force
 I = current
 ℓ = length
 P = power
 q = charge
 R = resistance
 r = separation
 t = time
 V = electric potential
 ρ = resistivity

WAVES

$$\lambda = \frac{v}{f}$$

f = frequency
 v = speed
 λ = wavelength

GEOMETRY AND TRIGONOMETRY

Rectangle
 $A = bh$
 A = area
 C = circumference
 V = volume

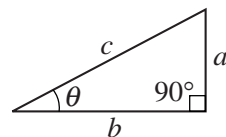
Triangle
 $A = \frac{1}{2} bh$
 S = surface area
 b = base
 h = height
 ℓ = length

Circle
 $A = \pi r^2$
 $C = 2\pi r$
 w = width
 r = radius

Rectangular solid
 $V = \ell wh$
Right triangle
 $c^2 = a^2 + b^2$

Cylinder
 $V = \pi r^2 \ell$
 $S = 2\pi r \ell + 2\pi r^2$
 $\sin \theta = \frac{a}{c}$
 $\cos \theta = \frac{b}{c}$
 $\tan \theta = \frac{a}{b}$

Sphere
 $V = \frac{4}{3} \pi r^3$
 $S = 4\pi r^2$



PHYSICS 1
Section I
40 Questions
Time—90 minutes

Note: To simplify calculations, you may use $g = 10 \text{ m/s}^2$ in all problems.

Directions: Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

1. In which of the following situations is the kinetic energy of the object decreasing?
 - (A) A sphere is dropped from a building.
 - (B) A satellite is moving in a circular orbit around Earth.
 - (C) A baseball is heading upward after being thrown at an angle.
 - (D) An elevator is moving upward at a constant velocity.

2. Two protons are held a distance d apart. The electrostatic force and the gravitational force that one proton exerts on the other are F_e and F_g , respectively. Which of the following correctly compares the magnitude and direction of these forces?

	<u>Magnitude</u>	<u>Direction</u>
(A)	$F_e > F_g$	Opposite
(B)	$F_e > F_g$	Same
(C)	$F_e < F_g$	Opposite
(D)	$F_e < F_g$	Same

3. A 2 kg object traveling at 5 m/s on a frictionless horizontal surface collides head-on with and sticks to a 3 kg object initially at rest. Which of the following correctly identifies the change in total kinetic energy and the resulting speed of the objects after the collision?

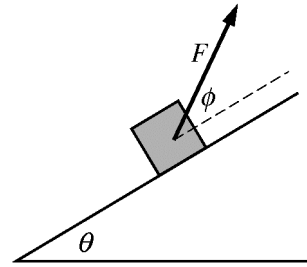
	<u>Kinetic Energy</u>	<u>Speed</u>
(A)	Increases	2 m/s
(B)	Increases	3.2 m/s
(C)	Decreases	2 m/s
(D)	Decreases	3.2 m/s

4. A stone of mass m is thrown upward at a 30° angle to the horizontal. At the instant the stone reaches its highest point, why is the stone neither gaining nor losing speed?
 - (A) Because the acceleration of the stone at that instant is zero
 - (B) Because the net force acting upon the stone at that instant has magnitude $m\vec{g}$
 - (C) Because the angle between the stone's velocity and the net force exerted upon the stone is 90°
 - (D) Because the stone follows a parabolic trajectory and the peak of the trajectory is where the parabola has zero slope

5. A small cart is rolling freely on an inclined ramp with a constant acceleration of 0.50 m/s^2 in the $-x$ -direction. At time $t = 0$, the cart has a velocity of 2.0 m/s in the $+x$ -direction. If the cart never leaves the ramp, which of the following statements correctly describes the motion of the cart at a time $t > 5 \text{ s}$?

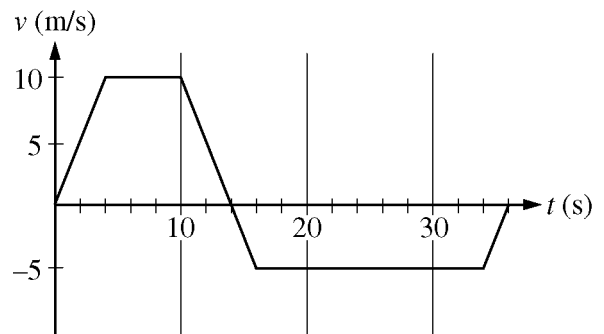
- (A) The cart is traveling in the $+x$ -direction and is slowing down.
- (B) The cart is traveling in the $+x$ -direction and is speeding up.
- (C) The cart is traveling in the $-x$ -direction and is slowing down.
- (D) The cart is traveling in the $-x$ -direction and is speeding up.

Item 6 was not scored.



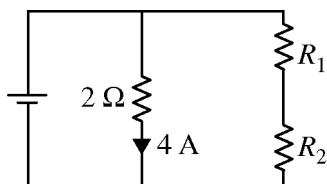
7. A box of mass m is on a rough inclined plane that is at an angle θ with the horizontal. A force of magnitude F at an angle ϕ with the plane is exerted on the block, as shown above. As the block moves up the plane, there is a frictional force between the box and the plane of magnitude f . What is the magnitude of the net force acting on the box?

- (A) $F \sin \phi - mg \cos \theta - f$
- (B) $F \cos(\phi + \theta) + mg \sin \theta - f$
- (C) $F \cos \phi - mg \sin \theta - f$
- (D) $F \cos(\phi + \theta) - mg \sin \theta - f$



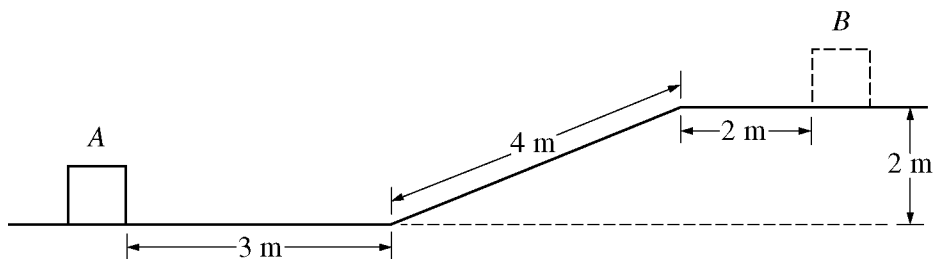
8. An object's velocity v as a function of time t is given in the graph above. Which of the following statements is true about the motion of the object?
- (A) The object is not moving from $t = 4 \text{ s}$ to $t = 10 \text{ s}$.
 - (B) The object's initial and final positions are the same.
 - (C) The object is slowing down from $t = 14 \text{ s}$ to $t = 16 \text{ s}$.
 - (D) The average acceleration of the object from $t = 0 \text{ s}$ to $t = 4 \text{ s}$ is different from the acceleration from $t = 34 \text{ s}$ to $t = 36 \text{ s}$.

Questions 9-10 refer to the following material.



In the circuit shown above, the sum of the resistances of resistors R_1 and R_2 is $8\ \Omega$.

9. What is the current through the battery?
- (A) 4 A
 - (B) 5 A
 - (C) 8 A
 - (D) 20 A
10. Resistor R_1 and the $2\ \Omega$ resistor are now swapped. How does the current in the right branch of the circuit change, and why?
- (A) The current does not change, because the total resistance does not change.
 - (B) The current increases, because the total resistance will always decrease.
 - (C) The current decreases, because the total resistance will always increase.
 - (D) The change in current cannot be determined without knowing the resistances of R_1 and R_2 .



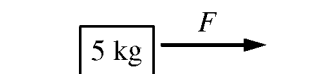
11. A block of mass 10 kg moves from position A to position B shown in the figure above. The speed of the block is 10 m/s at A and 4.0 m/s at B . The work done by friction on the block as it moves from A to B is most nearly
- (A) -280 J
 (B) -220 J
 (C) -200 J
 (D) 0 J
-
12. To determine the speed of waves on a string, some students tie a long string of unknown length between a wave generator and a wall. They vary the frequency f of the generator to get a standing wave. They count the nodes n and measure the wavelength λ . They repeat the experiment, creating standing waves with different frequencies. Which of the following is the best relationship to graph to determine the speed of the waves on the string?
- (A) f as a function of λ
 (B) f as a function of $1/\lambda$
 (C) f as a function of n
 (D) f as a function of $1/n$
13. A ladder at rest is leaning against a wall at an angle. Which of the following forces must have the same magnitude as the frictional force exerted on the ladder by the floor?
- (A) The force of gravity on the ladder
 (B) The normal force exerted on the ladder by the floor
 (C) The frictional force exerted on the ladder by the wall
 (D) The normal force exerted on the ladder by the wall
14. Some students want to calculate the work done by friction as an object with unknown mass moves along a straight line on a rough horizontal surface. The students have a force probe, a meterstick, and a stopwatch. Which of the following will allow the students to take the measurements needed to calculate the work done by friction?
- (A) Pulling the block at an unknown constant acceleration with the force probe for a measured time
 (B) Pulling the block at an unknown constant speed with the force probe for a measured time
 (C) Pulling the block at an unknown constant acceleration with the force probe for a measured distance
 (D) Pulling the block at an unknown constant speed with the force probe for a measured distance

15. A pendulum consisting of a sphere suspended from a light string is oscillating with a small angle with respect to the vertical. The sphere is then replaced with a new sphere of the same size but greater density and is set into oscillation with the same angle. How do the period, maximum kinetic energy, and maximum acceleration of the new pendulum compare to those of the original pendulum?

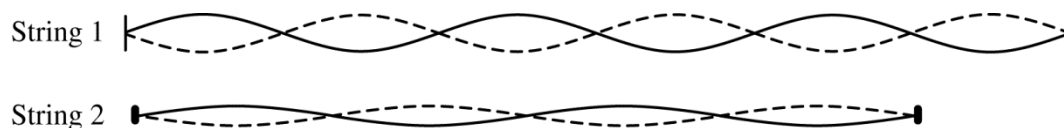
<u>Period</u>	<u>Maximum Kinetic Energy</u>	<u>Maximum Acceleration</u>
(A) Larger	Larger	Smaller
(B) Smaller	Larger	Smaller
(C) The same	The same	The same
(D) The same	Larger	The same

16. Planet X has twice Earth's mass and three times Earth's radius. The magnitude of the gravitational field near Planet X's surface is most nearly

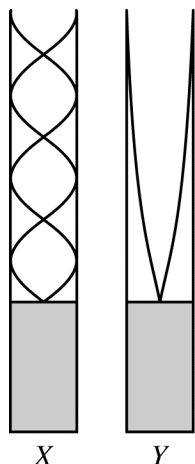
- (A) 2 N/kg
(B) 7 N/kg
(C) 10 N/kg
(D) 20 N/kg



17. A force F is exerted on a 5 kg block to move it across a rough surface, as shown above. The magnitude of the force is initially 5 N, and the block moves at a constant velocity. While the block is moving, the force is instantaneously increased to 12 N. How much kinetic energy does the block now gain as it moves a distance of 2 m?
- (A) 10 J
(B) 14 J
(C) 24 J
(D) 34 J

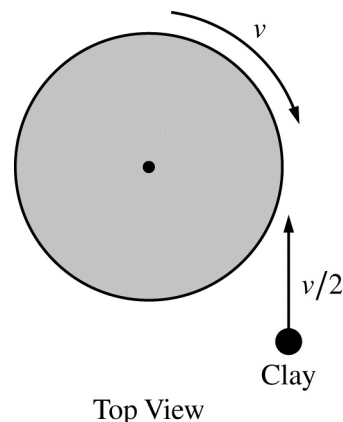


18. The figure above represents two guitar strings of different materials and lengths, which are on two guitars of different sizes. String 1 is plucked so it vibrates in the pattern shown. Very soon after string 1 is plucked, string 2, which is a short distance away, vibrates in the pattern shown. The guitars are placed in a sealed chamber and then the air is pumped out of the chamber. String 1 is again plucked and vibrates in the pattern shown. Does string 2 again vibrate in the pattern shown, and why or why not?
- (A) Yes, because waves again carry some of the energy produced by string 1 to string 2
 - (B) Yes, because the strings share the same fundamental frequency
 - (C) No, because the amplitude of the vibration of string 1 becomes zero too quickly for string 2 to start vibrating
 - (D) No, because almost no energy associated with the vibration of string 1 reaches string 2



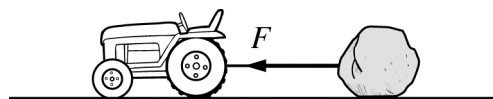
19. The figure above represents standing wave patterns in two identical tubes. The tubes contain the same amount of water, and the standing waves are produced by holding a vibrating tuning fork near the top of each tube. What is the relationship between the wavelengths λ_X and λ_Y of the standing waves?

- (A) $\lambda_X = \frac{1}{7}\lambda_Y$
- (B) $\lambda_X = \frac{2}{7}\lambda_Y$
- (C) $\lambda_X = \frac{7}{2}\lambda_Y$
- (D) $\lambda_X = 7\lambda_Y$



20. A system consists of a disk rotating on a frictionless axle and a piece of clay moving toward it, as shown in the figure above. The outside edge of the disk is moving at a linear speed v , and the clay is moving at speed $v/2$. The clay sticks to the outside edge of the disk. How does the angular momentum of the system after the clay sticks compare to the angular momentum of the system before the clay sticks, and what is an explanation for the comparison?
- (A) It is the same because there is no external torque acting on the system.
 - (B) It is greater because the rotating mass increases, which increases the rotational inertia.
 - (C) It is less because the speed of the disk decreases when the clay sticks to it.
 - (D) It is less because the angular momentum of the clay opposes that of the disk.

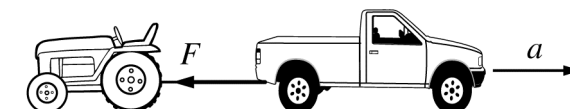
21. A disk of known radius and rotational inertia can rotate without friction in a horizontal plane around its fixed central axis. The disk has a cord of negligible mass wrapped around its edge. The disk is initially at rest, and the cord can be pulled to make the disk rotate. Which of the following procedures would best determine the relationship between applied torque and the resulting change in angular momentum of the disk?
- (A) Pulling on the cord, exerting a force of 15 N for 2 s and then 25 N for 3 s, and measuring the final angular velocity of the disk
- (B) For five different time intervals, pulling on the cord, exerting a force of 15 N, and then measuring the angle through which the disk rotates in each case
- (C) For five different time intervals, pulling on the cord, exerting a force of 15 N, and then measuring the final angular velocity of the disk
- (D) For five forces of different magnitude, pulling on the cord for 5 s, and then measuring the final angular velocity of the disk
22. When object X with charge $+2\ \mu\text{C}$ is 1 m from object Y with charge $-3\ \mu\text{C}$, the magnitude of the force between them is F . Object Y is removed, and object Z with charge $-2\ \mu\text{C}$ is placed 2 m from object X . What is the magnitude of the force between objects X and Z ?
- (A) $F/2$
- (B) $F/3$
- (C) $F/4$
- (D) $F/6$



Boulder: Force F to the left
No acceleration

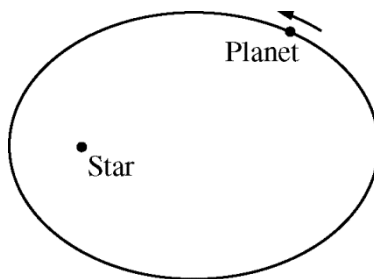


Wagon: Force F to the left
Acceleration a to the left

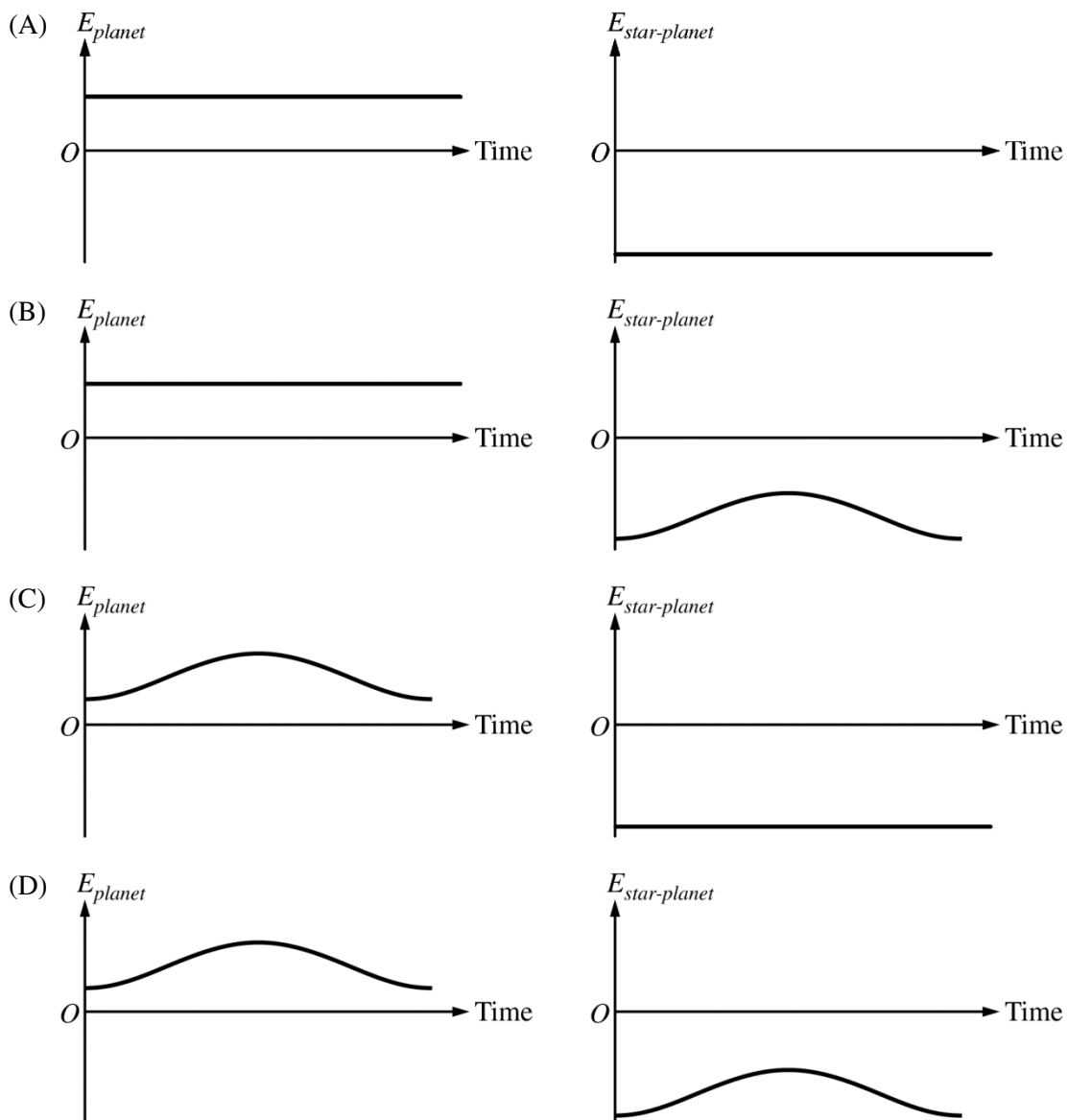


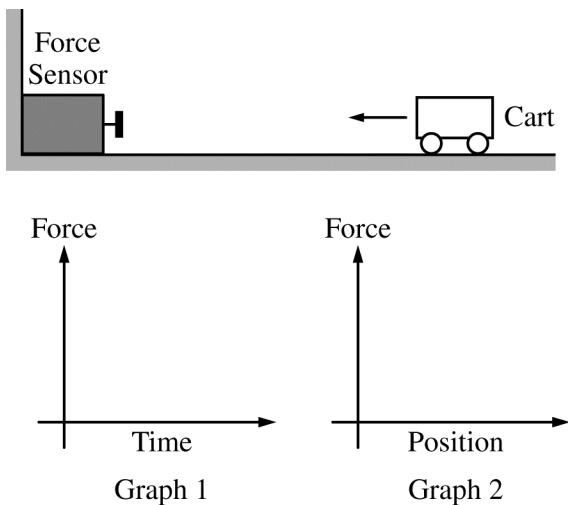
Truck: Force F to the left
Acceleration a to the right

23. Each of the figures above shows a tractor attached to an object. The tractor exerts the same constant force F on each object in every case. Which of the following is a true statement about an object and the relative magnitude of the force exerted by the object on the tractor?
- (A) The magnitude of the force exerted by the truck on the tractor is greatest, because the resulting motion is in the direction opposite the tractor's pull.
- (B) The magnitude of the force exerted by the boulder on the tractor is least, because no motion results.
- (C) The magnitude of the force exerted by the wagon on the tractor is least, because the resulting motion is in the direction of the tractor's pull.
- (D) The magnitude of the force exerted by each object on the tractor is equal, because the tractor exerts an equal force on each object.



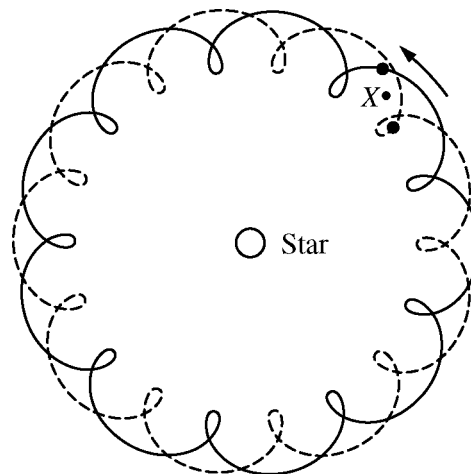
24. A planet is in an elliptical orbit around a star, as shown above. Which of the following best represents the mechanical energy E_{planet} of just the planet and the mechanical energy $E_{\text{star-planet}}$ of the star-planet system as functions of time for one complete orbit?





25. A cart of known mass moves with known speed along a level, frictionless track, as shown in the figure above. The cart hits a force sensor and rebounds. The force sensor measures the force exerted on the cart as a function of time and as a function of the position of the cart. The results will be graphed on the axes shown. Which of the two graphs can be used to determine the cart's speed after it rebounds?

(A) Only graph 1; graph 2 will have no information useful for finding the speed.
 (B) Only graph 2; graph 1 will have no information useful for finding the speed.
 (C) Either graph 1 or graph 2 can be used.
 (D) Neither graph alone is sufficient; both graph 1 and graph 2 are needed.

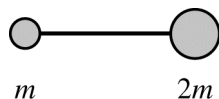


26. The figure above represents the orbits of two planets of equal mass that orbit their star in the counterclockwise direction as a double-planet system. From the point of view of an observer on either planet, the planets appear to orbit each other while also orbiting the star. The dots on the orbits represent the position of the planets at time t_0 , and X is the position of their center of mass at that time. Which of the following arrows best represents the acceleration of the center of mass of the double-planet system when it is at point X ?

- (A)
- (B)
- (C)
- (D)

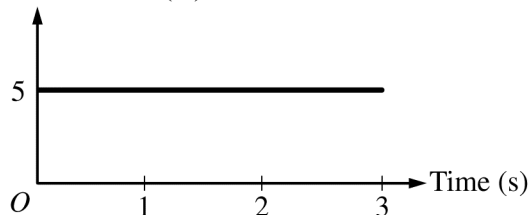
27. Two identical blocks are connected to the opposite ends of a compressed spring. The blocks initially slide together on a frictionless surface with velocity v to the right. The spring is then released by remote control. At some later instant, the left block is moving at $v/2$ to the left, and the other block is moving to the right. What is the speed of the center of mass of the system at that instant?
- (A) $5v/2$
(B) $3v/2$
(C) v
(D) $v/2$
28. A person holds a book at rest a few feet above a table. The person then lowers the book at a slow constant speed and places it on the table. Which of the following accurately describes the change in the total mechanical energy of the Earth-book system?
- (A) The total mechanical energy is unchanged, because there is no change in the book's kinetic energy as it is lowered to the table.
(B) The total mechanical energy is unchanged, because no work is done on the Earth-book system while the book is lowered.
(C) The total mechanical energy decreases, because the person does positive work on the book by exerting a force that opposes the gravitational force.
(D) The total mechanical energy decreases, because the person does negative work on the book by exerting a force on the book in the direction opposite to its displacement.

Questions 29-30 refer to the following material.



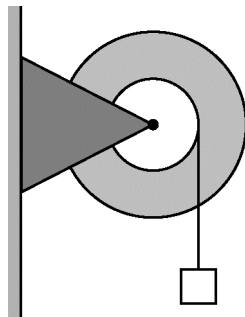
A system consists of two spheres, of mass m and $2m$, connected by a rod of negligible mass, as shown above. The system is held at its center of mass with the rod horizontal and released from rest near Earth's surface at time $t = 0$.

Rate of Change of
Linear Momentum of
the Sphere of Mass m (N)

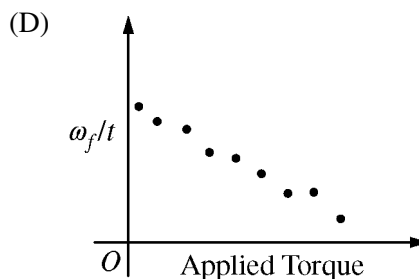
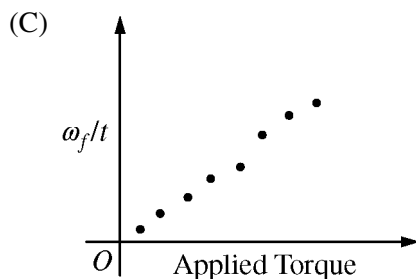
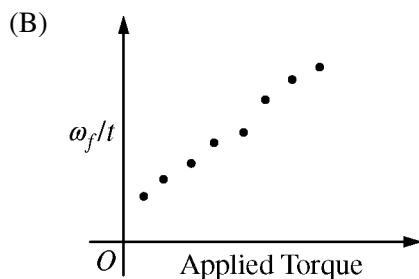
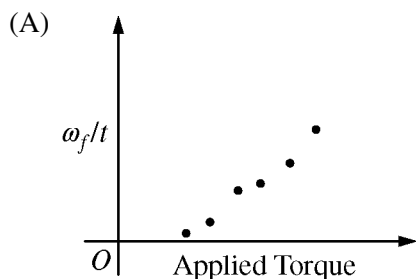


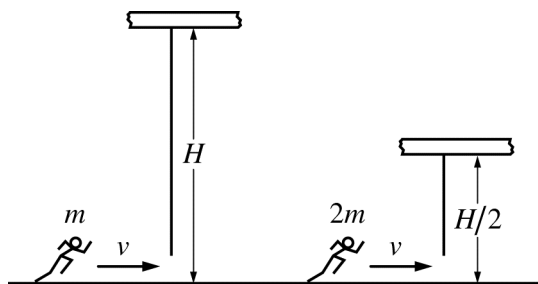
29. The graph above shows the rate of change of linear momentum of the sphere of mass m as a function of time. What is the linear momentum of the two-sphere system at time $t = 3.0$ s?
- (A) 5 kg•m/s
(B) 15 kg•m/s
(C) 45 kg•m/s
(D) 60 kg•m/s
30. Which of the following best explains why the system does not rotate around its center of mass as it falls?
- (A) The Earth exerts the same gravitational force on both spheres, causing them to accelerate at the same rate.
(B) The Earth exerts the same gravitational force on both spheres, generating torques that cancel out.
(C) The Earth exerts a larger gravitational force on the sphere of mass $2m$, but that sphere is closer to the center of mass and the torques cancel out.
(D) The Earth exerts a larger gravitational force on the sphere of mass $2m$, but that sphere has more inertia and the torques cancel out.

31. A person is running on a track. Which of the following forces propels the runner forward?
- (A) The normal force exerted by the ground on the person
 - (B) The normal force exerted by the person on the ground
 - (C) The force of friction exerted by the ground on the person
 - (D) The force of friction exerted by the person on the ground
32. In a one-dimensional perfectly elastic collision, an object of mass m is traveling with speed v_0 in the $+x$ -direction when it strikes an object with mass $3m$ that is at rest. What are the objects' velocities following the collision?
- | <u>Object of Mass m</u> | <u>Object of Mass $3m$</u> |
|--------------------------------------|---------------------------------------|
| (A) Zero | $v_0/3$, $+x$ -direction |
| (B) $v_0/4$, $+x$ -direction | $v_0/2$, $+x$ -direction |
| (C) $v_0/2$, $+x$ -direction | $v_0/2$, $+x$ -direction |
| (D) $v_0/2$, $-x$ -direction | $v_0/2$, $+x$ -direction |



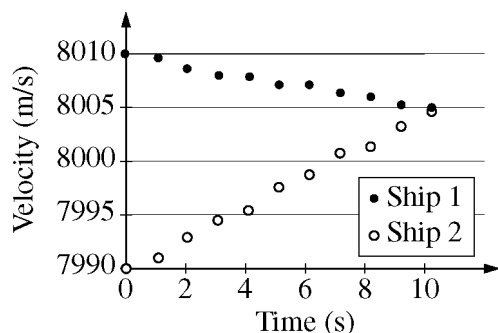
33. A student conducts an experiment to determine the relationship between applied torque and change in angular velocity. The student uses the apparatus shown in the figure above, consisting of two disks that are glued together and mounted on a horizontal axle. Blocks of varying mass are hung from a string wound around the smaller disk. The blocks are released from rest, exerting different torques on the disks, and are allowed to fall a fixed distance. For each block, the time of fall t and the final angular velocity ω_f of the disks are measured. There is considerable friction between the disks and the axle. Which of the following best represents a plot that can be obtained from the student's data?





34. An athlete with mass m running at speed v grabs a light rope that hangs from a ceiling of height H and swings to a maximum height of h_1 . In another room with a lower ceiling of height $H/2$, a second athlete with mass $2m$ running at the same speed v grabs a light rope hanging from the ceiling and swings to a maximum height of h_2 . How does the maximum height reached by the two athletes compare, and why?
- (A) The first athlete reaches a greater height, because this athlete swings on a longer rope.
- (B) The second athlete reaches a greater height, because this athlete has a greater mass.
- (C) The two athletes reach the same height, because the effect of the rope length offsets the effect of the athletes' masses.
- (D) The two athletes reach the same height, because the athletes run with the same speed.

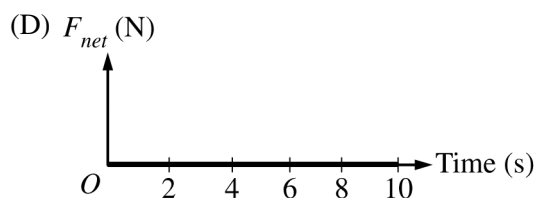
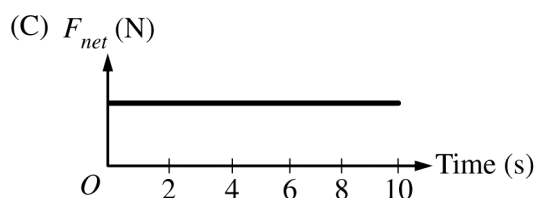
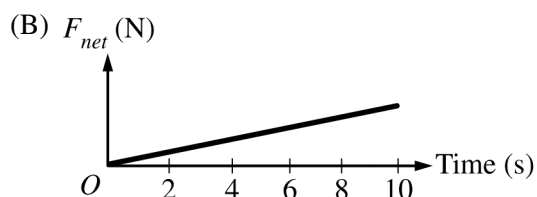
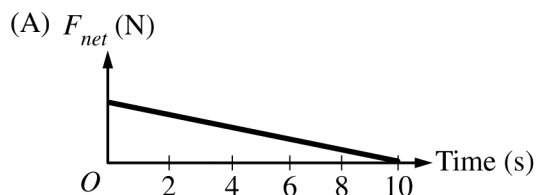
Questions 35-36 refer to the following material.



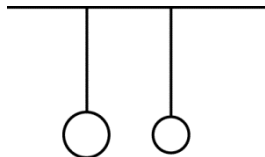
Two identical spaceships are traveling in deep space, far from any planets or stars. The ships travel in the same direction, with the slower one directly behind the faster one. The ships are connected by a cable attached to a spool, so that the part of the cable outside the ships can be made longer or shorter as needed. The cable is used to bring the ships to the same speed for a transfer of cargo. The graph above shows the speed of the two ships during a 10 s interval.

35. Does at least one of the ships have its engine turned on during the time interval shown, and what evidence indicates so?
- (A) Yes, because Ship 2 is speeding up.
 - (B) Yes, because the momentum of the two-rocket system increases.
 - (C) Yes, because an engine is needed to keep the system moving.
 - (D) No, because the cable alone could be responsible for making Ship 1 slow down and Ship 2 speed up.

36. Which of the following graphs best represents the net force F_{net} exerted on the two-ship system?

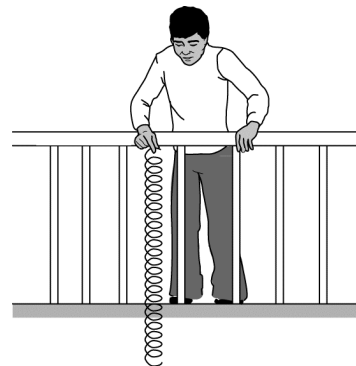


Directions: For each of the questions or incomplete statements below, two of the suggested answers will be correct. For each of these questions, you must select both correct choices to earn credit. No partial credit will be earned if only one correct choice is selected. Select the two that are best in each case and then fill in the corresponding circles that begin with number 131 on page 3 of the answer sheet.



131. Two hollow, uncharged conducting spheres hang by threads from the ceiling, as shown above. The spheres have the same mass but are different sizes. A charge $+Q$ is deposited on the larger sphere. The spheres are then momentarily brought into contact and separated, after which they move away from each other. Which of the following statements about the final state of the spheres and strings must be true? Select two answers.

(A) The spheres have charges of opposite sign.
 (B) The sum of the charges on the spheres is $+Q$.
 (C) The tension in each string is larger than it was initially.
 (D) The force exerted by the ceiling on each string is directed vertically upward.

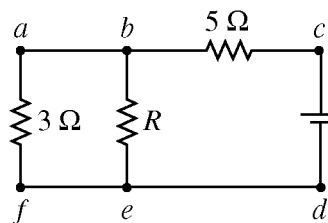


132. A student wants to demonstrate a transverse wave for a friend. The student holds a long spring with his fingertips and lets it hang vertically over the edge of a railing, as shown above. The student can move his hand in a way that creates waves that propagate down the spring. Which of the following movements of the student's hand will create a transverse wave? Select two answers.

(A) Left and right
 (B) Forward and backward
 (C) Up and down
 (D) Releasing the spring

133. Suppose two square wave pulses that are in the same plane could be created on a string, one with a maximum displacement of 4 cm and the other with a maximum displacement of 2 cm. Which of the following is a possible displacement of the string when the wave pulses overlap? Select two answers.

(A) 2 cm
(B) 3 cm
(C) 6 cm
(D) 8 cm



134. The figure above shows a circuit with three resistors—two with known resistance and one with unknown resistance—and a battery with unknown potential difference. Which of the following sets of measurements of potential difference will allow determination of the unknown resistance R ? Select two answers.

(A) Between points *b* and *e* only
(B) Between points *a* and *f* and between points *b* and *c*
(C) Between points *b* and *c* and between points *c* and *d*
(D) Between points *a* and *f* and between points *b* and *e*

PHYSICS 1
Section II
5 Questions
Time—90 minutes

Directions: Questions 1, 4 and 5 are short free-response questions that require about 13 minutes each to answer and are worth 7 points each. Questions 2 and 3 are long free-response questions that require about 25 minutes each to answer and are worth 12 points each. Show your work for each part in the space provided after that part.



1. (7 points, suggested time 13 minutes)

Cart A has mass M and is released from rest at a height $2H$ on a ramp making an angle 2θ with the horizontal, as shown above. Cart B has mass $2M$ and is released from rest at a height H on a ramp making an angle θ with the horizontal. The carts roll toward each other, have a head-on collision on the horizontal portion of the ramp, and stick together. The masses of the carts' wheels are negligible, as are any frictional or drag forces.

- (a) Indicate whether the carts remain at rest, move to the left, or move to the right after the collision.

_____ Remain at rest _____ Move to the left _____ Move to the right

Explain how you arrived at your answer.

(b) Consider the time interval from when the two carts are released until just after they collide.

- i. For the system consisting of the two carts and Earth, indicate whether the total mechanical energy increases, decreases, or remains the same.

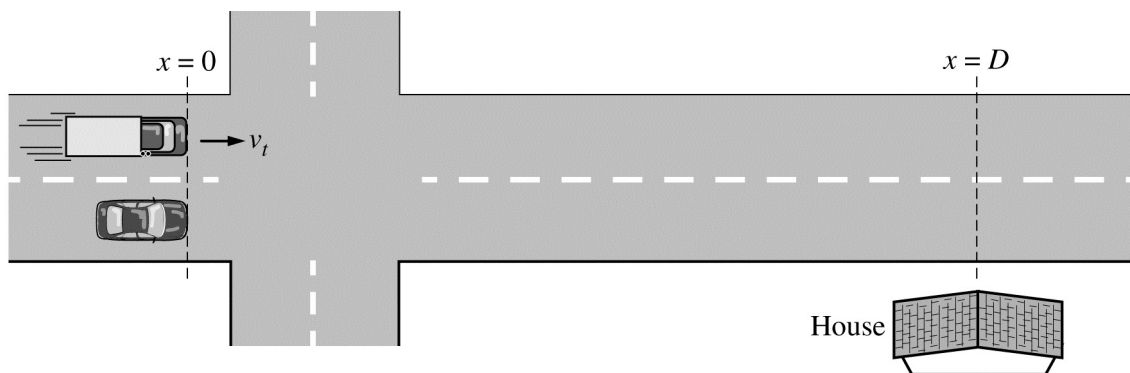
____ Increases ____ Decreases ____ Remains the same

Justify your answer.

- ii. For the system consisting of only the two carts, indicate whether the total mechanical energy increases, decreases, or remains the same.

____ Increases ____ Decreases ____ Remains the same

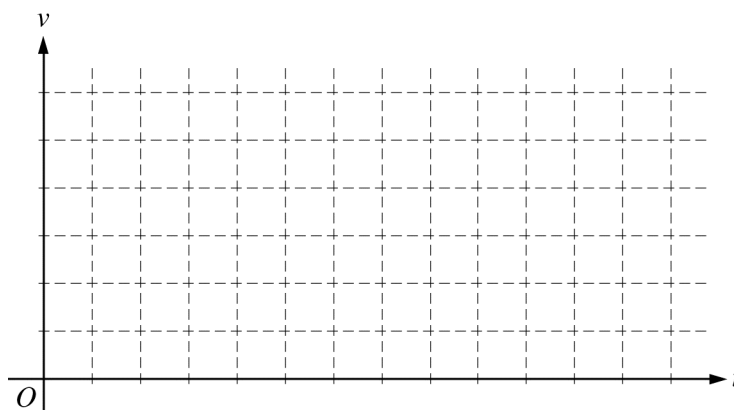
Justify your answer.



2. (12 points, suggested time 25 minutes)

A car is stopped at a traffic light. The light turns green, and at time $t = 0$ the car starts moving and travels with a constant acceleration. At that instant a truck traveling at constant speed v_t is alongside the car, with the front of each vehicle at position $x = 0$, as shown above. The truck passes the car, but the car later catches up to the truck in front of a house, such that at time t_D the front of each vehicle is at position $x = D$.

- (a) On the axes below, sketch and label graphs of the velocity of the car and the velocity of the truck as a function of time. Indicate any important velocities or times.



- (b) Two students are discussing how the speed of the car compares to the speed of the truck when both vehicles are in front of the house.

Student 1 says, "The distance traveled by the car and the truck is the same, and the time is the same, so they must have the same speed."

Student 2 says, "I don't see how that can be. The car catches up to the truck, so the car has to be going faster."

- i. Which aspects of Student 1's reasoning, if any, are correct? Support your answer in terms of relevant features of your graphs in part (a).

- ii. Which aspects of Student 2's reasoning, if any, are correct? Support your answer in terms of relevant features of your graphs in part (a).

- (c) Derive an expression for the acceleration of the car. Express your answer in terms of D and v_t .

- (d) Determine the time at which the speed of the car is equal to the speed v_t of the truck. Express your answer in terms of t_D . Justify your answer.

3. (12 points, suggested time 25 minutes)

You are given a long length of string and an oscillator that can shake one end of the string at any desired frequency. The oscillator has a display that indicates the frequency. You are asked to design an experiment to study how the velocity of waves on the string depends on the string's tension. You do not have any way to measure time with sufficient accuracy to help in your investigation.

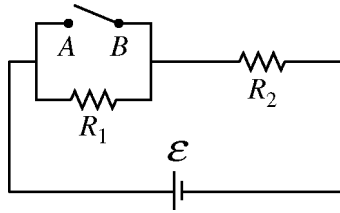
(a) Describe your experimental setup and procedure, including any additional pieces of equipment you would need and the kind of data you would record. Include enough detail that another student could follow and complete the experiment successfully.

(b) Describe how you would analyze your data to obtain information about the wave velocity's dependence on tension.

- (c) A student performs the experiment and can see from the data table below that the velocity of the waves on the string increases when the tension in the string increases. The student claims that the velocity is directly proportional to the tension. Do the results support the student's claim? Explain in detail how you arrived at your answer.

Velocity (m/s)	17	24	29	34	38
Tension (N)	2.0	4.0	6.0	8.0	10.0

- (d) After the experiment is completed, a student attaches a string to the wall and the other end of the string to an oscillator. The frequency of the oscillator is adjusted to make a standing wave. The string is allowed to vibrate in a standing wave with constant amplitude.
- While the string is vibrating as a standing wave with constant amplitude, is the oscillator doing work on the string? Explain your reasoning.
 - While the string is vibrating as a standing wave with constant amplitude, is the string gaining mechanical energy? Explain your reasoning.



4. (7 points, suggested time 13 minutes)

The circuit shown above contains two resistors, an ideal battery, and a switch. The resistance of resistor R_1 is less than the resistance of resistor R_2 .

- (a) Indicate whether the current through resistor R_2 when the switch is open is greater than, less than, or equal to the current through resistor R_2 when the switch is closed.

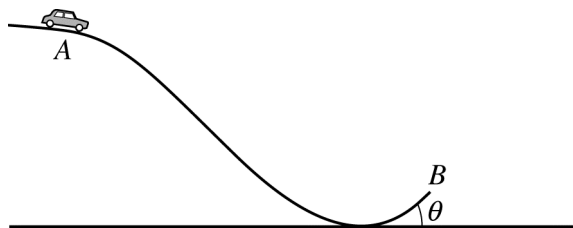
_____ Greater than _____ Less than _____ Equal to

Briefly explain how you arrived at your answer.

- (b) Closing the switch creates a short circuit. Indicate whether the absolute value of the potential difference between points A and B when the switch is open is greater than, less than, or equal to the absolute value of the potential difference when the switch is closed.

_____ Greater than _____ Less than _____ Equal to

Explain your reasoning in a clear, coherent paragraph-length explanation that may also contain equations and/or drawings. Address the conditions before and after the switch is closed.



5. (7 points, suggested time 13 minutes)

A toy car coasts along the curved track shown above. The car has initial speed v_A when it is at point A at the top of the track, and the car leaves the track at point B with speed v_B at an angle θ above the horizontal. Assume that the rotational kinetic energy of the car's wheels and energy losses due to friction are both negligible.

(a) Suppose the toy car is released from rest at point A ($v_A = 0$).

- i. After the car leaves the track and reaches the highest point in its trajectory it will be at a different height than it was at point A . Briefly explain why this is so.
- ii. Determine the speed of the car when it is at the highest point in its trajectory after leaving the track, in terms of v_B and θ . Briefly explain how you arrived at your answer.

(b) Suppose the toy car is given an initial push so that it has nonzero speed at point A . Determine the speed v_A of the car at point A such that the highest point in its trajectory after leaving the track is the same as its height at point A . Express your answer in terms of v_B and θ . Explain how you arrived at your answer.

Answer Key for AP Physics 1 Practice Exam, Section I

Question 1: C	Question 21: D
Question 2: A	Question 22: D
Question 3: C	Question 23: D
Question 4: C	Question 24: C
Question 5: D	Question 25: C
Question 6: *	Question 26: A
Question 7: C	Question 27: C
Question 8: B	Question 28: D
Question 9: B	Question 29: C
Question 10: D	Question 30: C
Question 11: B	Question 31: C
Question 12: B	Question 32: D
Question 13: D	Question 33: A
Question 14: D	Question 34: D
Question 15: D	Question 35: B
Question 16: A	Question 36: C
Question 17: B	Question 131: B, C
Question 18: D	Question 132: A, B
Question 19: A	Question 133: A, C
Question 20: A	Question 134: B, C

*Item 6 was not scored.

AP[®] PHYSICS

2015 SCORING GUIDELINES

GENERAL NOTES ABOUT 2015 PHYSICS SCORING GUIDELINES

1. The solutions contain the most common method of solving the free-response questions and the allocation of points for this solution. Some also contain a common alternate solution. Other methods of solution also receive appropriate credit for correct work.
2. Generally, double penalty for errors is avoided. For example, if an incorrect answer to part (a) is correctly substituted into an otherwise correct solution to part (b), full credit will usually be awarded. One exception to this may be cases when the numerical answer to a later part should be easily recognized as wrong, e.g., a speed faster than the speed of light in vacuum.
3. Implicit statements of concepts normally receive credit. For example, if use of the equation expressing a particular concept is worth one point, and a student's solution contains the application of that equation to the problem but the student does not write the basic equation, the point is still awarded. However, when students are asked to derive an expression it is normally expected that they will begin by writing one or more fundamental equations, such as those given on the exam equation sheet. For a description of the use of such terms as “derive” and “calculate” on the exams, and what is expected for each, see “The Free-Response Sections—Student Presentation” in the *AP Physics; Physics C: Mechanics, Physics C: Electricity and Magnetism Course Description* or “Terms Defined” in the *AP Physics 1: Algebra-Based and AP Physics 2: Algebra-Based Course and Exam Description*.
4. The scoring guidelines typically show numerical results using the value $g = 9.8 \text{ m/s}^2$, but use of 10 m/s^2 is of course also acceptable. Solutions usually show numerical answers using both values when they are significantly different.
5. Strict rules regarding significant digits are usually not applied to numerical answers. However, in some cases answers containing too many digits may be penalized. In general, two to four significant digits are acceptable. Numerical answers that differ from the published answer due to differences in rounding throughout the question typically receive full credit. Exceptions to these guidelines usually occur when rounding makes a difference in obtaining a reasonable answer. For example, suppose a solution requires subtracting two numbers that should have five significant figures and that differ starting with the fourth digit (e.g., 20.295 and 20.278). Rounding to three digits will lose the accuracy required to determine the difference in the numbers, and some credit may be lost.

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Question 1

7 points total

**Distribution
of points**

(a) 3 points

For any application of conservation of energy

1 point

For correctly applying conservation of energy for each cart

1 point

For any application of conservation of momentum

1 point

Example 1: The contributions of each cart to the potential energy of the carts-Earth system when the carts are at their points of release are the same, because one has twice the height and one has twice the mass of the other. All this potential energy is converted into kinetic energy when the carts reach the bottom of the ramp, so the carts have the same kinetic energy there. This means that the speed of cart A is 1.4 times the speed of Cart B. Before the collision, cart A has a momentum of about $1.4Mv$ to the right; cart B has a momentum of $2Mv$ to the left. The net momentum is to the left, so after the collision the carts move left.

Example 2:

$$Mg(2H) = Mv_A^2/2 ; \quad v_A^2 = 4gH ; \quad v_A = 2\sqrt{gH}$$

$$2Mg(H) = (2M)v_B^2/2 ; \quad v_B^2 = 2gH ; \quad v_B = \sqrt{2gH}$$

$$M(2\sqrt{gH}) - 2M(\sqrt{2gH}) = (M + 2M)v_f = 3Mv_f$$

$$v_f = \left(\frac{2 - 2\sqrt{2}}{3} \right) \sqrt{gH} = -\frac{0.83}{3} \sqrt{gH} = -0.28\sqrt{gH}$$

One earned point is deducted for any incorrect use of ramp angles.

(b)

i. 2 points

For indicating that no energy is gained as the carts move down the ramps, or using the result of the reasoning in part (a) regarding the energy for this part of the motion

1 point

For indicating that energy is dissipated in the collision and a comparison of initial and final energies consistent with the students' assumptions about energy before the collision

1 point

ii. 2 points

For indicating that the initial mechanical energy of the carts is zero

1 point

For a comparison with the energy after the collision consistent with previous responses

1 point

Example: When the carts are released, they have no mechanical energy, since they are not moving. After the collision, the carts have kinetic energy. The total mechanical energy of the cart-cart system increases.

Alternate solution

Alternate points

For identifying gravitational force as the only external force doing work

1 point

For correct use of the work-kinetic energy theorem

1 point

Example: For the system of just the two carts, there is no initial mechanical energy, since the carts are at rest. The gravitational force exerted on the carts by Earth is the only external force doing work, and the work on the system is positive. So the kinetic energy of the system increases, which increases the total mechanical energy.

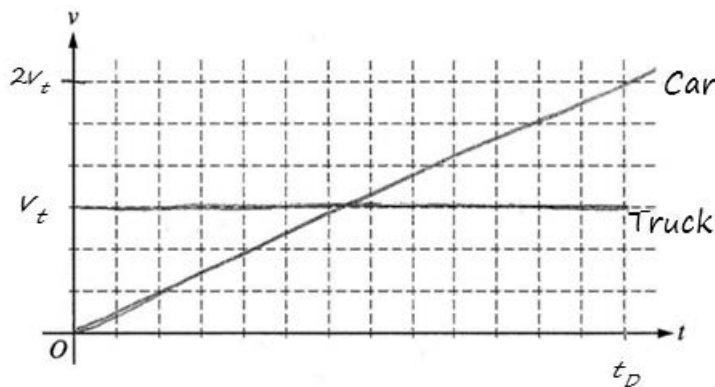
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Question 2

12 points total

**Distribution
of points**

(a) 3 points



For a clearly labeled horizontal line for the truck

1 point

For a clearly labeled straight line with positive slope for the car

1 point

For the speed of the car ending higher than the speed of the truck at a time labeled to indicate that the house has been reached

1 point

(b)

i. 2 points

For indicating that student 1 is correct that the car and truck travel the same distance in the same time

1 point

For identifying a feature of the graph that supports this: e.g., the area under both curves is the same, or the visual average of the speed of the car is the same as the speed of the truck

1 point

ii. 2 points

For indicating that student 2 is correct that the car's speed is greater when the vehicles are in front of the house

1 point

For identifying a feature of the graph that supports this: e.g., the final speed of the car is twice the speed of the truck

1 point

(c) 3 points

For indicating that the final speed of the car is twice the speed of truck

1 point

For applying any other appropriate kinematic relationship(s)

1 point

$$(2v_t)^2 = v_{c0}^2 + 2aD$$

For solving for the acceleration in terms of D and v_t

1 point

$$a = 2v_t^2/D$$

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Question 2 (continued)

**Distribution
of points**

(d) 2 points

For indicating that the car and truck have the same speed at $t = t_D/2$

1 point

For a correct justification

1 point

Examples:

That's where the graphs intersect.

At t_D the car's speed is twice the truck's speed, so they'll be equal at $t_D/2$.

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Question 3

12 points total

**Distribution
of points**

- (a) 3 points
- For describing a method for setting and changing the tension in the string (e.g., hanging the string over a pulley and attaching an object whose mass has been measured, then using objects of different mass) 1 point
 - For using the oscillator to create a wave on the string, adjusting the frequency of the oscillator or the length of the string for each tension until a standing wave is established 1 point
 - For describing a method for measuring quantities relevant to determining the wavelength of the standing wave formed on the string (e.g., measuring the length of the string with a meter stick and counting the number of nodes) 1 point
- (b) 3 points
- For describing a correct method of calculating the propagation speed of the wave which includes either measurement or calculation of the wavelength of the standing wave 1 point
 - For describing a correct method of calculating or measuring tension (e.g., when using objects of different mass, calculate $F_T = mg$) 1 point
 - For describing a correct method of analyzing the relationship between the wave velocity and tension (e.g. graph velocity as a function of tension) 1 point
- (c) 2 points
- For a reasonable attempt to check whether velocity and tension are directly proportional 1 point
 - For a valid justification leading to the conclusion that velocity and tension are not directly proportional (e.g., best-fit line for a plot of the data does not go through zero; doubling tension does not result in doubled propagation speed) 1 point
- (d) i. 2 points
- The oscillator is doing work on the string.
 - For indicating that oscillator generates the wave by applying a force to the string over a distance 1 point
 - For indicating that work is the product of the applied force and the distance over which it is applied 1 point
- Alternate solution* *Alternate points*
- For indicating that resistive forces will take energy away from the string* 1 point
 - For indicating that if the amplitude is constant the oscillator is supplying energy by doing work on the string* 1 point

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Question 3 (continued)

**Distribution
of points**

(d) (continued)

ii. 2 points

The string is not gaining mechanical energy.

For indicating that the energy carried by a mechanical wave is related to its amplitude,
and having no incorrect or irrelevant statements

1 point

For indicating that the wave does not gain mechanical energy because its amplitude is
constant

1 point

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Question 4

7 points total

**Distribution
of points**

(a) 2 points

The correct answer is “Less than”.

For a correct justification leading to a correct conclusion

2 points

Example: The current through R_2 when the switch is open is less than when the switch is closed, because the total resistance of the circuit with the switch closed is less.

Note: One point can be earned for a reasonable but incorrect justification that is correctly used.

Examples:

The current through R_2 is the same when the switch is open or when the switch is closed, because regardless of whether the switch is open or closed, any current that is split between R_1 and AB must combine again as it moves through R_2 .

The current through R_2 when the switch is open is greater than when the switch is closed, because closing the switch adds an additional path for current flow.

(b) 5 points

When the switch is open:

For indicating that there is a potential difference across R_1 because the full current through the battery flows through the resistor

1 point

For indicating that the potential difference across the switch is equal to the potential difference across R_1

1 point

When the switch is closed:

For indicating that the resistance of the switch is zero or negligible, or assuming the switch has resistance and indicating that the resistance of the parallel combination of the switch and resistor is less than that of the resistor alone

1 point

For indicating implicitly or explicitly that the potential difference across the switch is zero (or negligible), because all (or some) current goes through the switch that has no (or little) resistance and zero (or little) current goes through R_1

1 point

For a complete explanation having no incorrect or physically irrelevant statements and indicating that the potential difference across the switch is greater when the switch is open

1 point

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Question 5

7 points total

**Distribution
of points**

- (a)
- i. 2 points
- For indicating that the mechanical energy of the car-Earth system is constant between point A and the highest point in the car's trajectory 1 point
- For indicating that the car is still moving at its highest point and has some kinetic energy so the car-Earth system must have less gravitational potential energy, therefore must be at a lower height than it was at point A 1 point
- ii. 2 points
- For indicating that the velocity of the car at its highest point is equal to the horizontal speed at point B 1 point
- For stating that the speed at the highest point is $v_B \cos \theta$ 1 point
- (b) 3 points
- For implicitly or explicitly applying conservation of energy 1 point
- For implicitly or explicitly indicating that the gravitational potential energy is the same at the beginning and at the end 1 point
- For indicating that the speed is the same at the beginning and at the end, so $v_A = v_B \cos \theta$, or consistent with the answer in part (a-ii) 1 point